

● HARD

● INFORMATION AND IDEAS

● ANSWER KEY

Inferences

05

10 answers · Rationale-rich · Teach from doc

Q1**D**

- D. Mars's crust likely formed as a result of other major geological events in addition to the cooling of a magma ocean.
-

Choice D is the best answer. Cooling magma would create basalt, but “a planetary surface that formed in a mostly basaltic environment would be unlikely to contain large amounts of silica.” Since Mars's crust does contain large amounts of silica, it is unlikely that Mars's crust was formed exclusively by cooling magma. Therefore, there were likely other major geological events that created the high silica concentrations.

Choice A is incorrect. Although the passage discusses these two methods of collecting data about Mars's crust, it never compares their reliability, so there's no basis for this inference.

Choice B is incorrect. The passage never mentions anything about the crusts of other planets, so there's no basis for this inference. Choice C is incorrect. The passage never mentions Earth's crust, so there's no basis for this inference.

Q2

C

- C. human activity likely played no role in the introduction of the sweet potato in Polynesia.
-

Choice C is the best answer because it most logically completes the text's discussion of the sweet potato in Polynesia. The text indicates that the sweet potato is found in Polynesia but originated in South America, and that the sweet potato was being cultivated by Native Hawaiians and other Indigenous peoples in Polynesia long before sea voyages between South America and Polynesia began. The text goes on to note that research by Muñoz-Rodríguez and colleagues has established that the Polynesian varieties of sweet potato split from South American varieties more than 100,000 years ago, which is thousands of years before humans settled in Polynesia. If Polynesian peoples were cultivating the sweet potato before sea voyages between Polynesia and South America began, and if Polynesian varieties of sweet potato diverged from South American varieties well before people were in Polynesia, it can reasonably be concluded that humans didn't play a role in bringing the sweet potato to Polynesia.

Choice A is incorrect. The text doesn't provide any information about when the sweet potato began to be cultivated in South America, so there's no support for the conclusion that cultivation began in Polynesia before it began in South America. Choice B is incorrect because the text indicates that the sweet potato was being cultivated in Polynesia long before sea journeys between Polynesia and South America began. Therefore, it wouldn't be reasonable to conclude that Polynesian peoples acquired the sweet potato from South American peoples. Additionally, the text indicates that the Polynesian varieties of sweet potato diverged from the South American varieties thousands of years before people settled in Polynesia, which suggests that the sweet potato was already present in Polynesia when people arrived. Choice D is incorrect because the text states that the domestic sweet potato, which is found in Polynesia, descends from a wild South American plant, not from a domesticated South American plant. The only people that the text describes as cultivating the sweet potato are Native Hawaiians and other Indigenous peoples of Polynesia.

Q3

B

- B. some Ancestral Puebloans migrated to the Rio Grande Valley in the late 1200s and carried farming practices with them.
-

Choice B is the best answer because it presents the conclusion that most logically follows from the text's discussion of Ancestral Puebloans' migration to the Rio Grande Valley. The text states that in the late 1200s C.E., the Ancestral Puebloan civilization abandoned villages in its original homeland, which included the Mesa Verde site. The text goes on to say that recent genetic analysis has demonstrated that the modern turkey population in the Rio Grande Valley descends partly from the ancient turkeys raised at Mesa Verde, and that the genetic markers shared by the two turkey populations first appeared at Mesa Verde only after 1280 C.E. Therefore, it can reasonably be concluded that some Ancestral Puebloans migrated to the Rio Grande Valley in the late 1200s and carried their agricultural practices—including the farming of turkeys—to their new home.

Choice A is incorrect because the text never compares the condition of the Rio Grande Valley's terrain to that of Mesa Verde's terrain, either in the present or in the past. Choice C is incorrect. Although genetic analysis has demonstrated that the modern turkey population in the Rio Grande valley descended in part from the turkey population raised by the Ancestral Puebloans of Mesa Verde before their migration to the valley in 1280, this finding doesn't eliminate the possibility that Indigenous peoples living in the valley before 1280 might also have farmed turkeys. Choice D is incorrect. The text doesn't consider the possibility that before their migration to the Rio Grande Valley after 1280, the Ancestral Puebloans of Mesa Verde might have adopted turkey farming from an outside Indigenous civilization in another region; instead, the text provides evidence suggesting that the Ancestral Puebloans brought turkey farming to another region—the Rio Grande Valley—after 1280.

Q4

C

- C. *S. caninervis* growing under quartz crystals experience lower light intensity and are thus able to retain more moisture.
-

Choice C is the best answer because it most logically completes the text. The text explains that while desert moss species need sufficient sunlight for photosynthesis, exposure to sunlight can also affect the plants negatively by drying them out. Ekwealor and Fisher's team found specimens of *S. caninervis* moss growing under quartz crystals that were semitransparent, allowing some but not all sunlight to pass through them, and the shoot tissue of these specimens was longer than that of *S. caninervis* specimens found growing on the surface that were unprotected by such crystals. Since, as the text explains, the length of shoot tissue is an indicator of plant growth, and since greater exposure to sunlight results in greater loss of moisture, it can be inferred that *S. caninervis* growing under quartz crystals experience lower light intensity and are thus able to retain more moisture.

Choice A is incorrect because the text doesn't mention another moss species besides *S. caninervis* or discuss whether other such species are able to grow under transparent crystals, as *S. caninervis* is. Choice B is incorrect because, as the text explains, specimens of *S. caninervis* were found growing under quartz crystals and exhibited more plant growth than specimens growing on the soil surface. This wouldn't have been the case if the crystals transmitted insufficient light for the moss's photosynthesis. Choice D is incorrect. The text contrasts the growth of *S. caninervis* specimens found beneath semitransparent quartz crystals with that of specimens found on the soil surface absent such crystals, but it doesn't make a comparison among specimens growing beneath crystals of different degrees of thickness or consider how the relative thickness of the crystals affects the growth of the moss.

Q5

B

- B. rate the listening experience as more negative than those whose listening experience was uninterrupted.
-

Choice B is the best answer. It most logically completes the text. The text tells us that disruptions to an enjoyable experience increase viewers' enjoyment. It also says that researchers suspect the opposite is true for disruptions to an unpleasant experience. Thus, we can infer that the researchers expect to find that the interrupted unpleasant experience was worse for listeners than the uninterrupted unpleasant experience.

Choice A is incorrect. It doesn't logically complete the text. The text never makes any claims about how irritating the disruptions themselves are perceived to be. Rather, the text says that pleasant experiences are perceived as more enjoyable with interruptions, and that the opposite is suspected to be true of unpleasant experiences. Choice C is incorrect. It doesn't logically complete the text. The text never makes any claims about how long any experience is perceived to be. Rather, the text says that pleasant experiences are perceived as more enjoyable with interruptions, and that the opposite is suspected to be true of unpleasant experiences. Choice D is incorrect. It doesn't logically complete the text. The text never makes any claims about how interruptions affect the perceived volume of the unpleasant or pleasant experience. Rather, the text says that pleasant experiences are perceived as more enjoyable with interruptions, and that the opposite is suspected to be true of unpleasant experiences.

Q6

C

C. is probably in a location Isla hadn't ever come across while on the ground.

Choice C is the best answer because it most logically completes the text's discussion of Johnny Isla and the whalelike geoglyph. The text indicates that the German exhibit about the Nazca Lines included a photograph showing a whalelike geoglyph that Isla hadn't known about before attending the exhibit, even though Isla "specializes in" Nazca Lines geoglyphs. Given his expertise, and his surprise at being unfamiliar with the whale glyph, the text strongly suggests that Isla believed he would have noticed the glyph if he had been to its location. Thus, the text implies that the whalelike geoglyph is likely in a location Isla had not previously been to.

Choice A is incorrect because the text doesn't address either the species of whale that the geoglyph is meant to represent or its relationship to the earliest humans in the area that is now Peru. Choice B is incorrect. Although the text indicates that the photograph of the whalelike geoglyph was on display at a "German exhibit," that exhibit was specifically "about the Nazca Lines," which the text indicates are located in Peru. Choice D is incorrect. Although the text does indicate that the glyphs were created "over a period of many centuries," the text doesn't address when in that period of time any particular glyphs were created.

Q7

A

- A. while its content largely predates the invasion, *Cantares Mexicanos* also contains additions made after the invasion.
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Choice A is the best answer because it most logically completes the text. The text explains that the *Cantares Mexicanos* contains poems about the Aztec Empire from before the Spanish invasion. Furthermore, it indicates that notes in the collection attest that some of these poems predate the Spanish invasion, while some customs depicted are likely Spanish in origin. The implication is that some poems were composed before the invasion but the references to Spanish customs could have come about only after the invasion, and thus that the collection includes content that predates the invasion and also content from after the invasion.

Choice B is incorrect because the text clearly indicates that the collection is in Nahuatl, not Spanish, so the compilers' unfamiliarity with Spanish is irrelevant to whether the collection contains material composed after the Spanish invasion. Choice C is incorrect because the text mentions only the Aztec Empire and Spain: there is no information about the relationship of Aztec literature to any traditions other than its own or Spain's. Choice D is incorrect because the text states that some of the poems make "inarguable references" to common Spanish customs, which conflicts with the idea that these references can reasonably be attributed to mere coincidence.

Q8

B

- B. may promote increases in marsh plants or decreases in marsh plants, depending on the crabs' location.
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Choice B is the best answer. The text says that crab burrowing in the pannes enables plants to grow there again. It also says that crab burrowing at the edges of the marsh speeds up marsh loss. This suggests that burrowing crabs can either help or hurt marshes, depending on where they're located.

Choice A is incorrect. This inference isn't supported. The text never discusses pannes of different sizes. Choice C is incorrect. This inference isn't supported. The text never suggests that crabs do more burrowing in the pannes (the areas with standing water) than they do at the edges. Rather, the text says that the burrowing that happens in the pannes is beneficial, while the burrowing that happens at the edges is harmful. Choice D is incorrect. This inference isn't supported. The text never discusses the intensity of crab burrowing, nor does it discuss the size of the panne relative to the size of the marsh.

Q9

A

- A. some disagreements between the model's simulations of Titan's precipitation and humidity and the moon's actual precipitation and humidity are to be expected.
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Choice A is the best answer because it most logically completes the text's discussion of modeling the atmosphere of Titan. The text establishes that the methane mole fraction (the ratio of methane to other atmospheric components) influences precipitation, humidity, and other meteorological phenomena and that observational data for Titan are too lacking ("too sparse and discrepant") to fully determine the range of this measure across the moon's various atmospheric levels. The text indicates that because of this limitation, researchers developed an atmospheric model (IPSL) that uses a simplified approach: it applies a uniform methane mole fraction for the lowest atmospheric level. Since it is very unlikely that the actual methane distribution across atmospheric levels on Titan is uniform, and since the methane mole fraction directly affects precipitation and humidity, it logically follows that the model's simplified treatment of the methane mole fraction would lead to some disagreements between the model's simulations and Titan's actual precipitation and humidity.

Choice B is incorrect because the text makes it clear that the model *already* uses a single value for Titan's methane mole fraction instead of a range. Further, the text suggests that the single value is a simplification of what is likely a more complex reality, so there's no reason to assume that a single value would be more desirable than a range. Choice C is incorrect. The text strongly suggests that Titan's actual methane mole fraction differs from the uniform value used in the model, even though some of the model's outputs align closely with observations, but nothing in the text indicates how the actual value would differ. There's no reason to assume that the actual methane mole fraction is higher than what the model uses. The "lowest level" mentioned in the text refers to the atmosphere and not Titan's methane mole fraction. Choice D is incorrect because the text makes it clear that the model *doesn't* reflect the actual variations in the methane mole fraction across Titan's atmospheric levels, since observational data are too limited. Because the model uniformly applies a set methane mole fraction, no inconsistencies in the simulations would be attributable to actual variations in the moon's methane mole fraction.

Q10

A

A. bipedalism evolved because it was advantageous to a tree-dwelling ancestor of humans.

Choice A is the best answer because it most logically completes the text's discussion of the evolution of bipedalism in humans. According to the text, one potential explanation for humans walking upright on two legs is that the behavior evolved from an ancestor that mostly stayed on the ground and walked on four limbs, as modern chimpanzees and gorillas do. However, the finding that orangutans, also a relative of humans, sometimes stand on two legs in trees while using their arms to balance and reach for fruits suggests another possible explanation: perhaps a tree-dwelling ancestor of humans began moving on two legs because it offered an advantage, such as access to certain foods.

Choice B is incorrect because the finding that modern orangutans (a relative of humans) sometimes stand on two legs in trees doesn't offer any insight into when either bipedalism or tree-climbing behavior emerged in human ancestors. Additionally, the text indicates that one theory is that bipedalism evolved from a mostly ground-based ancestor that was already practicing knuckle-walking, not that bipedalism and knuckle-walking developed at the same time. Choice C is incorrect because the finding that orangutans (a relative of humans) sometimes stand on two legs in trees doesn't offer any insight into how difficult it would've been to move between the ground and the trees without bipedalism; there's no suggestion that climbing or moving in trees depends on the ability to walk on two legs rather than four, even if that ability might be helpful in certain circumstances. Choice D is incorrect because the finding that orangutans (a relative of humans) sometimes stand on two legs in trees doesn't suggest that a knuckle-walking human ancestor could've easily moved on two legs in trees. Although the text indicates that bipedalism may have evolved from a human ancestor that mostly stayed on the ground and walked on four limbs, it gives no indication of how easy it would've been for such an ancestor to move bipedally in trees.

Notes

Accept equivalent forms (e.g. $0.25 \equiv 1/4$). For grid-in items, decimal and fraction answers are both valid.

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